Experiencing and Envisioning Food: Designing for Change – Bonacho et al. (Eds)
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Making data digestible

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ABSTRACT: Physicalising data affords new kinds of interaction that open opportunities for meaning making. In our research, we consider how using food as the material for data physicalisation might expand the impact of this emerging field of practice in relation to Food System transformation. Food is sensorially rich, culturally and politically potent, and environmentally impactful; its resonance may be felt keenly at a range of scales from the personal and situated, to the systemic and global. To examine the impact of using food to construct data, we discuss three examples: i) the launch of a public Food Lab; ii) a temporary installation focused on food waste and sustainability; and iii) a short design research masters project on food and sustainable futures. Across these cases food acts as icebreaker, prompt for new thinking and sustenance, as well as a potent vehicle for design experimentation. In examining them, we unfold the ways that using food as data may make data more digestible, and thus more impactful, for different contexts and actors.

1 GENERAL INTRODUCTION

1.1 *Food system data*

Data is an inseparable part of people's lives. Every day, large amounts of data are produced, captured, transferred, transformed, and stored by individuals, institutions, organisations and nations the world over. Our research is concerned with the Human Food System, and the challenge of raising awareness about its social and environmental impacts, towards systemic change. The human food system is responsible for 20-40% of greenhouse gas emissions in every country and worldwide (Rosenzweig et al. 2020). Adding to this impact, by 2050 the world's food demand is expected to increase by 60% due to urbanization and continuous population growth (Alexandratos & Bruinsma 2012). Furthermore, food insecurity is a growing concern – in 2021, around 2,3 billion people were moderately or severely food insecure (FAO et al. 2022). These numbers indicate significant environmental and societal challenges that must be addressed if we are to have a fair and just food system for all.

Food System data is readily accessible on the internet, including through social media, where NGOs use social marketing to expand their reach (Wisetsri et al. 2021). The impact of these, often ephemeral, information encounters is, however, unclear.

Data must be accessible and understandable if it is to be impactful (Huron et al. 2014) – if it is to resonate with people, so that they take note and, from there, take action. This move from knowledge to action is key to societal change and, like food system transformation, must unfold individually and systemically.

To open up thinking about how to expand the impact of food system data, we turn to the emerging practice of data edibilization (Wang et al. 2016) – constructing physical manifestations of data using food. We consider if making food system data edible might make it more digestible, such that it prompts critical reflection on food system transformation and, from there, prompts people to be in action.

We use three contexts to explore these ideas: i) the launch of a Food Living Lab; ii) a temporary installation about food waste and sustainability; and iii) a three-week design research master's project on food and sustainable futures. Through this work, we make tentative steps towards expanding the impact of data edibilization towards being a tool for societal change-making. The research is part of FUSILLI[1], a four-year Horizon 2020 project whose aim is to foster the transformation of the food system in 12 European cities.

1 https://fusilli-project.eu

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Kolding, Denmark, where this research takes place, is one of the twelve cities.

2 EDIBLE MEANING-MAKING

2.1 *From physicalisation to edibilisation*

It is almost a decade since Jansen et al., (2015) coined the term data physicalisation for the practice of constructing physical manifestations of data. Data physicalisation has since been explored through artistic representatiions using environmental data (Pigott et al. 2022); for novel engagement with highly personal data (Panagiotidou et al. 2020); and to serve as instruments to be applied to everyday problem solving (Jansen et al. 2015). These explorations, and others, demonstrate many advantages of making data tangible. For instance, when experienced in 3D forms, graphs become more memorable for people and lead to greater impact of their experience of the data. Adding interactivity to physicalisations has been shown to further enhance people's capacities for reflection and meaning-making (Karyda et al. 2021).

A wide range of materials are used for physicalisations. Examples include using sound to sonify data (Barrass 2016); developing 3D graphs (Jansen et al. 2015) and interactive and static data sculptures (Kosminsky & de Oliveira 2021) using everyday materials such as beads and iron wires; creating living physicalisations, using plants (Botros et al. 2016) and more. Only recently, food has been use in systematic and intentional ways for physicalising data, with the results described as data edibilisations (Wang et al. 2016; Mueller et al. 2021).

2.2 *The potential of edible data*

Using food as a medium for physicalising data shows great potential. Food affords new kinds of interaction with data that open up new opportunities for meaning making. Data cuisine[2] is an early example of data edibilisations that was developed outside of research. Curator, Susanne Jaschko and data visualisation expert, Moritz Stefaner, developed a series of workshops that bring together chefs and communities to create edible physicalisations of data that is important to them. The project specifically focuses on exploring the language of food for communicating information. The range of approaches they use vary from representational graphs in tangible form, to more abstract representations such as a dish cooked from ingredients gathered within a specific time frame.

In other examples technology can be core to data edibilization. "Cyber Wagashi" uses wasabi to represent weather data ('CYBER WAGASHI | OPEN MEALS' n.d.), bridging two kinds of technology – a satellite data platform and a 3D food printer – to physicalise the data. The aim of yber Wagashi is to make people "feel" the climate of the day by tasting it. Similarly, Tasty Beats (Khot et al. 2015) physicalises heartbeat data through consumable drinks, allowing participants to combine data and exercise in a pleasurable experience.

Despite these rich and intriguing examples, food remains an underexplored medium for physicalising data as an inseparable part of everyday life practices. We hope to expand this area of concern. Wang et al., (2016) describe data edibilizations as affective, rich, memorable and social. Mueller et al., (2021) identify four themes that guide design engagements with food: form, commensality, ephemerality and emotional response. They describe engagement with data-food as a multisensory affective experience with great potential. We build on their work to take data edibilisation into the realm of food system transformation.

3 THE THREE CASES

We explore data edibilisation in three contexts: i) the launch of a public Food Lab; ii) a temporary public installation focused on food waste and sustainability; and iii) a short (three-week) design research masters project on food and sustainable futures. Each context affords new ways of understanding data due to the ways that food was used to represent the data, as well as the kinds of interactions the physicalisations afforded. We now describe the three cases.

3.1 *Food lab launch*

Food Lab is a food living lab for community-based experimentation and co-creation of sustainable food futures[3]. Food Lab is based in Kolding, Denmark, and was launched November 2021. For the launch, we invited local food stakeholders, including politicians, municipal representatives, educators, entrepreneurs, farmers, representatives of the water and energy companies, a rewilding consultancy, a poet and more.

Any good launch involves food. Food Lab launch was no exception. It included a buffet of edible data, prepared using rescued surplus foods (Figure 1). Both the data and the foods were provided by Food ReFormers – a local NGO who focus on food waste and surplus[4].

Food Reformers run a community fridge from the front garden of the building where Food Lab is situated. The fridge is refilled daily with surplus foods rescued from local supermarkets before it is thrown into dumpsters. The food is within its use-by date, but does not have enough time left

2 https://data-cuisine.net/

3 https://koldingfood2030.dk/index.php/food-lab-2/

4 https://www.facebook.com/foodreformerhttps://www.facebook.com/foodreformer



Figure 1. Four data edibilisations prepared for the food lab launch. ⏎

to make it suitable for sale. At the community fridge, people can help themselves to this food any time of day or night. Within six months of its use, the community fridge saved 6,3 tons of surplus food that was being disposed of by local supermarkets.

Our buffet represented four datasets. The aim was to prompt discussion around the local food system:

a) ***more-than-food*** was a pile of pastries and bread (Figure 1a). In Denmark, these foods serve for social rather than nutritional needs.
b) ***don't forget your roots*** was a beetroot and carrot soup made from foods of unknown origin, collected from supermarkets without packaging (Figure 1b). According to the community fridge, around 40% of the food that they collect is without packaging, and thus lacks information on product nutrition and origin.
c) ***food survivors*** represented the amount of food saved by donations in Denmark yearly (Figure 1c).
d) ***travel distance*** physicalised the country of origin of each ingredient of a guacamole and the distance travelled to arrive at our table, with labelling and strings that expand out from the bowl of food. The distances the ingredients travelled was physically scaled and took up significant space on the table (Figure 1d).

Compared to the corpus of data edibilisation (see for example (Mueller et al. 2021)), the food was consumed by individuals who had a stake in the data, and who were at the event to reflect on how food is treated locally. The physicalisation, therefore, aims to not only prompt discussion but to also influence decision making. While we have no evidence that it contributed directly to decision-making, on the night it prompted vibrant discussions among key decision-makers about food waste, food practices, sustainability, and food system transformation challenges.

3.2 *International Park(ing) Day*

Our second case took place during Park(ing) Day, an annual "global, public, participatory art project" in which "people across the world temporarily repurpose street parking spaces and convert them into tiny parks and places for art, play, and activism."[5]

On Park(ing) Day 2021 in Kolding, FoodReFormers installed a dining table for five, across two parking spots outside Food Lab. Similar to the Food Lab launch, the installation used rescued surplus food to physicalise data related to this food. The set-up included a dining table with five "dishes" (Figure 2). A menu card next to each dish introduced relevant statistics.

The physicalisations were in most cases quite literal, so as to be easy to digest by a broad, self-selecting audience. The aim was to foster discussion with random passersby.

Dishes included:

a) A selection of cold cuts in a tall vase, representing the number of animals slaughtered every year – the diversity represented through different coloured packaging (Figure 3a). This dish highlighted that 20% of global meat production is wasted annually (Karwowska et al. 2021).
b) A kilo of bread (Figure 3b). The accompanying menu card complemented the representation by showing that a kilo of wasted bread equals 1600 L of wasted water.
c) an empty dish to communicate food waste reduction (Figure 3c).
d) five small spice containers, three on the plate, two off (Figure 3d). This dish was meant to physicalise a 68% decline in global vertebrate species populations between 1970 and 2016. (WWF 2020).
e) three packs of crackers, one sitting outside the dish to indicate that 1/3 of food is being wasted worldwide[6] (Figure 3e).

5 https://www.myparkingday.org/
6 https://www.saveonethird.org/



Figure 2. Food Reformers installation on Park(ing) Day. ⏎

classes of approximately twenty students have undertaken the project since 2019. We describe the class then discuss two examples.

The class takes inspiration from an article by The Centre for Genomic Gastronomy[7] (2015), and uses unrelated statistics as the basis for experiments. Students form small groups and imagine a desirable future world, in 2050 – not too far from today, but far enough to combine imaginaries of things from the present and things from the future. Reimagining the social is critical in this process, and is done from a range of perspectives. As explained by Jasanoff & Kim (2015), social imaginaries – collective beliefs about how society functions – can enable or disable societal transformation and are critical to its realisation. Their



Figure 3. a) Five data edibilisations prepared by Food ReFormers for Park(ing) Day. ⏎

This small intervention was part of an ongoing discussion that Food ReFormers is having with different publics in Kolding. On the day, representatives from FoodReFormers and FUSILLI (author 3 & 2, respectively) stayed with the table for several hours inviting locals to join them in discussion. In general, people seemed curious: interested to see what was on the table and to listen to the reasonings behind each dish, though responses were mixed. Clearly, more sophisticated data edibilisations that invited embodied engagement may have been more successful. However, that may not have been the only factor at play.

Food System change is not straightforward. Food is culture and identity. We eat to express who we are, connect to others, learn and express our social and cultural alliances, and stimulate the gustatory system and the senses, as well as for nutrition (Douglas 1972; Warde 2016; Warde & Martens 2000; Ochs & Shohet 2006). Changing the food system therefore may require that people change their understanding of who they are, their culture, habits and practices.

3.3 *Food and sustainable futures*

Our third case relates to a three-week Design Masters project on Food and Sustainable Futures. Six

world-making is conducted from the perspectives of: a human from the present; a human from 2050 (who could not be alive today); a living non-human; and a human-made thing. Students employ tools, symbols and processes of food preparation to experimentally expand their thinking.

The Centre for Genomic Gastronomy article (2015) explains that artists working with food can ask *what if* in three ways: *diegetic, figurative* and *realistic*. We provide their description, reframed towards the students:

a) Following the *diegetic* approach the students could create props and images of food that cannot be eaten. The aim is to express something about food or the future without being constrained by material reality, the laws of physics or biology.
b) Following the *figurative* approach, students develop metaphors using food that can be eaten, but that does not have to taste good or be nutritious. This approach positions the audience as active participants, and confronts them with the reality of consuming something.
c) In the *realistic* approach, foods from the future can be served for human consumption today. Such foods use novel ingredients, (bio)technologies, and cooking processes to implicate the

7 https://genomicgastronomy.com

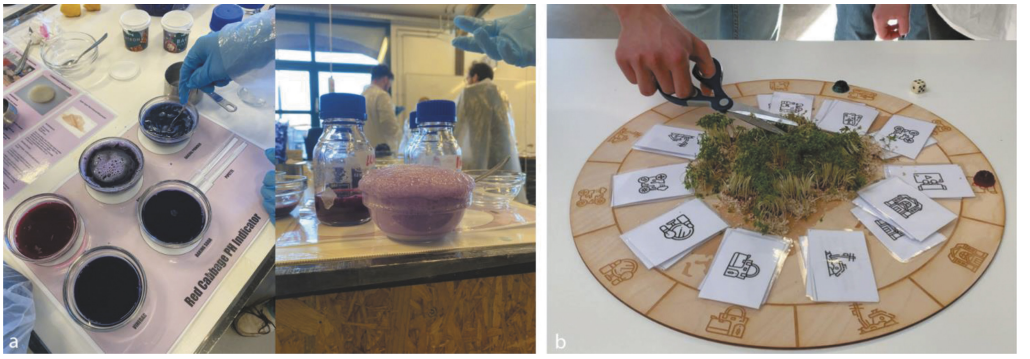


Figure 4. a) material experiments for 'Edible Immigration Levels'; b) cutting an alfalfa sprout forest in 'True Cost'. ⏎

eater directly in the culture of the imagined alternative food culture.

We ask the students to experiment with these possibilities as designers, and in the process to consider what art might bring to design, as well as what design might bring to this particular kind of artistic thinking and working. To exemplify the students' experiments we unpack two examples that demonstrate how changes in material properties or framing may prompt meaning-making.

Edible Immigration levels (Figure 4a): During material experiments, a group of students combined bicarbonate soda with lemon juice and purple cabbage extract. The combination resulted in a surprisingly explosive material event that prompted all who saw it to pay close attention. Inspired by the dynamic materiality of the event, the students paired this with their data to tell a story about the explosive dynamic of immigration which will be triggered by rising sea levels. They used the materials – the food – while telling their story to trigger the frothy explosion. The food was then served to their classmates and the taste of it triggered nuanced reflections about the narrated story. Intriguingly, the food tasted slightly different for each person and thus prompted different associations.

True Cost (Figure 4b): In our second example, students created a two-player board game that was presented 'during a break in the 2050 World Sustainability Conference'. As the class entered the room where the presentation took place, they were welcomed to the break and thus immediately understood their role. The game includes a round board, a dice and two counters, a small forest of alfalfa sprouts in the centre of the board, a pile of cards corresponding to each square on the board, and a pair of scissors. Players role the die. When they land on a square, they select the corresponding card and read out a dilemma – a choice between something perhaps prosaic and something intriguing or desirable that has a significant carbon cost. Depending on their choice the corresponding amount of alfalfa sprouts is cut from the small forest on the table. We thus literally see the true carbon cost of their choice.

Despite only two students playing the game, the set-up of the conference break was convincing, and everyone was engaged from the outset, cheering on the players, howling with dismay when they made an unsustainable choice, and lamenting the loss of the forest. From the perspective of data edibilisation, the food made the narrative tangible and thereby shifted the way that the audience related to what was at stake.

4 DISCUSSION & CONCLUSION

Our cases use food in a variety of ways. In social contexts the food acts an *ice-breaker*, helping people to find focus for the social encounter. By supporting casual engagement with data, the representations literally become food for thought, engaging the sensory to become a richly loaded prompt for discussion: *a prompt for new thinking*. Where the food is edible, it may also serve as *sustenance*, as in the Food Lab launch. When the representation is in the form of an edible sculpture, or reframed as a forest, it defamiliarises assumptions around the data and thereby becomes *a ticket-to talk*. Beyond these possibilities, when designers experiment with food as a bio-design material, it may serve as *a potent vehicle for design experimentation,* and the properties and interactions as inspiration for both representation and storytelling.

The two first cases provide the reader with real life scenarios of using data edibilisation to prompt critical thinking about food practices. At the food lab launch, physicalising the data prompted people to reflect on their practices while they were eating the food. A number of participants at the event have since requested data edibilisations for future events focused on food system transformation.

At the Park(ing) day installation, the recontextualised dining table prompted reflection through looking, rather than engaging in an embodied way. The data was not edible, but if we look to the three ways of asking what if provided by the Centre for Genomic Gastronomy (2015), the food does not need to be edible to be impactful.

Our third case offers two examples that bring focus to the material properties of food and how this might be leveraged to prompt critical reflection on challenging issues, such as forced immigration, or the environmental impact of our practices. Both cases involved embodied engagement with the food – the first through watching an explosive event then eating the result; the second by cutting it back, as if killing a forest. Both cases connect to powerful data which is rendered potent through its transformation into data edibilisations.

We argue that foregrounding food as culture when designing edible data physicalisations is significant. The highly personal, socio-cultural resonances of the food influences how people understand the physicalisations and may thereby prompt new kinds of reflection. The mouth feel and taste of the food, literally influences the thinking process. Storytelling through food-based data physicalisations affords unique understandings because taste is a highly personal, idiosyncratic process. And while not explored in depth in this article, we propose that embodied engagement is fundamental to edibilisations having lasting impact. On these grounds, we suggest that developing physicalisations with food, in particular when cultural considerations ar leveraged, expands the possibilities of meaning-making in rich, impactful, and heretofore unforeseen ways.

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